Normal Distribution of the 18th hole

Introduction

Golf is very important to me. I was eight years old when my dad took me to my first golf tournament. Soon after, I began to play the sport that has quickly become a favorite. Almost every part of me can be seen on the golf course, my determination has been reinforced on the course, my patience has been strengthened with each shot, but most importantly, my interests have stemmed from my 14 clubs and Callaway golf balls. Therefore, it was quite evident that if I were to dive deeper into the crevices of an area that involved math, it would be with golf.

The tournament that I have the closest connection to is the Northern Trust Open. For the last 6 years every mid-February, my family pays a visit to world renowned Riviera Country Club to witness more golf history being made. In years previous, I have come home with autographs from world number ones, pictures with caddies, new drills to use at practice, but my favorite has, by far, been the prized 18th hole ball that the winner hands to one lucky person in the crowd. In 2016 that was me.

At golf courses around the world, the 18th hole is known by many, usually for its toughness or maybe for the relief that the good or bad round is finally over. To me, the 18th hole is the most relaxing feeling because I know that after I get the ball into the hole, I get to grab my free chocolate milkshake the tournament provides. However that is not the case for the professionals, in fact the 18th hole may be the most important hole of their round. So, how crucial is this hole exactly? At the Northern Trust Open, the par 4 is 451 yards and is a 4/10 handicap that many people can possibly have trouble with. As one of golf's most famous finishing holes, the legendary 18th has defined Riviera's challenge for more than 80 years. The primary test lies in the tremendously intimidating tee shot, a semi-blind drive played to a fairway hidden beyond a steep, Kikuyu-covered hillside. From there, a mid to long iron is generally needed to reach a stable, no bunker putting surface, which is nestled memorably in a natural amphitheater beneath the clubhouse. The hole, simple to some, but treacherous to others, is a classic.

Aim:

With money riding on all eighteen holes, but immensely on the last hole, what is deemed as a normal distribution of the shots taken on the last hole? Are the scores abundantly pars or does the pressure consume a golfer and higher scores are a result? In that manner, what will the distribution of the 18th hole scores be from the Northern Trust Open played at Riviera Country Club?

Investigation:

Golf has been played for centuries and so the 18th hole has been played countless times, as well. Specifically, the Northern Trust Open tournament has been held for the past 91 years, but the most recent years have brought among stars of all countries and can prove essential to understanding if there is a normal distribution of scores at the famous 18th hole. Essentially, to acquire the most accurate data possible, multiple years have to be looked at. Therefore, Rounds 1 and 2 where all golfers compete before the cuts from years 2012 to 2016 will help to allude to how scores are normally distributed. On the contrary, if they are not, how is "normal distribution" determined. Lastly, with the data from the previous 5 years, I can predict what the data will potentially depict for 2017.

2012:

From February 16 to February 19, 144 golfers participated in the tournament taking place at Riviera Country Club, but 3 players did not compete in day 2 of the 4-day tournament. Thus, the data will be taken from 141 golfers.

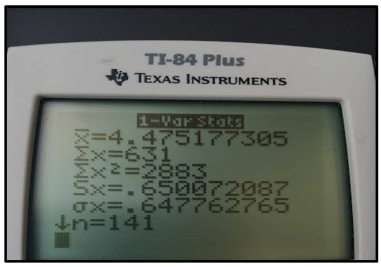
Round 1

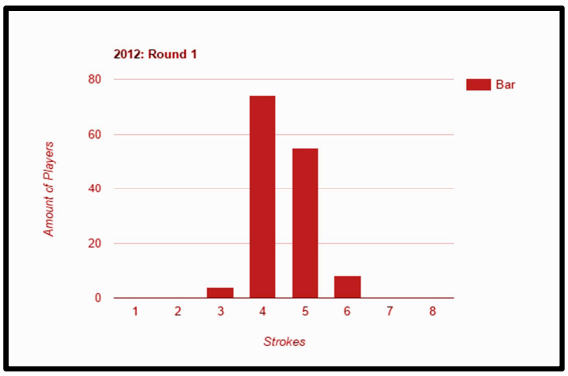
Hole 18- Par 4

X	1	2	3	4	5	6	7	8
P (X=x)	$\frac{0}{141}$	$\frac{0}{141}$	$\frac{4}{141}$	$\frac{74}{141}$	$\frac{55}{141}$	$\frac{8}{141}$	$\frac{0}{141}$	$\frac{0}{141}$

E (x)= $1(\frac{0}{141}) + 2(\frac{0}{141}) + 3(\frac{4}{141}) + 4(\frac{74}{141}) + 5(\frac{55}{141}) + 6(\frac{8}{141}) + 7(\frac{0}{141}) + 8(\frac{0}{141}) = \frac{631}{141} = 4.48$

σ=0.648

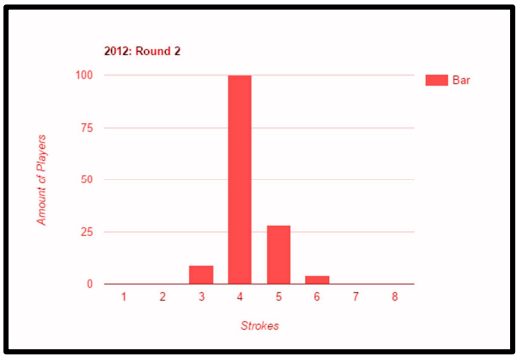




Round 2

Hole 18- Par 4

X	1	2	3	4	5	6	7	8
P (X=x)	$\frac{0}{141}$	$\frac{0}{141}$	$\frac{9}{141}$	$\frac{100}{141}$	$\frac{28}{141}$	$\frac{4}{141}$	$\frac{0}{141}$	$\frac{0}{141}$



E (x)= $1(\frac{0}{141}) + 2(\frac{0}{141}) + 3(\frac{9}{141}) + 4(\frac{100}{141}) + 5(\frac{28}{141}) + 6(\frac{4}{141}) + 7(\frac{0}{141}) + 8(\frac{0}{141}) = \frac{591}{141} = 4.19$

σ=0.582

Analysis:

From the 2012 data, the expected value, also called the mean, depicts that Round 1 had an average of 4.48 shots while Round 2 had an average of 4.19. Additionally, the standard deviation was found. This is beneficial to depict, as it is the measure of the dispersion of a set of data from its mean. Essentially, a normal distribution follows a specific pattern: one standard deviation percentage is 68%, two standard deviation percentages is 95%, and 3 is close to 100%. In Round 1, one standard deviation to the right would amount to 5.128 and to the left it would be 3.832. The scores that would fit in this deviation would be the whole number scores, 4 and 5, so $\frac{129}{141}$ (91%) of the golfers were in this range.

Reflection:

This percentage is visibly much higher than the normal distribution percentage of 68% of one standard deviation to the right and left. While two standard deviations to the right would be at

5.776 and to the left would be 3.184. This contains 95% of the values. This would still only encompass scores of 4 and 5, but the 91% would be closer to the typical percentage. Lastly, 3 standard deviations to the left and right would contain be a value of 6.424 on the right and 2.536 on the left. The scores would then expand to 3, 4, 5, and 6 strokes being $\frac{141}{141}$(100%) being alike to the 3 standard deviation right and left percentage of a normal distribution curve. This was the first example to seeing if the 18th hole data would be normally distributed.

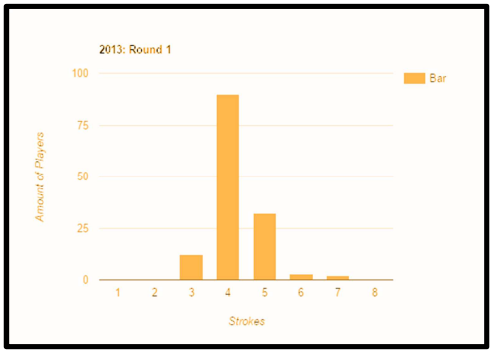
Analysis:
For Round 2, the expected value was 4.19 and the standard deviation was 0.582. With one standard deviation to the right and left, the values were respectively 4.772 and 3.608. The scores would then only contain the golfers who had 4 swings at the ball. This would be$\frac{100}{141}$(71%) which is fairly close to a normal distribution curve percentage of 68%. Next, two standard deviations are 5.354 and 3.026 which would include the golfers with 4 and 5 strokes taken on the hole. This percentage is $\frac{128}{141}$(91%) which is also relatively close to the normal distribution curve. Lastly, three standard deviations the value are 5.936 and 2.444 which includes 3, 4, and 5 strokes. The percentage would be $\frac{137}{141}$(97%) which is proximately like the normal distribution curve.

Reflection:
Evidently, this makes the data taken represent a different type of curve as the data is not exactly alike to a normal distribution, but is barely off in many cases. Likewise, this can be inferred for all the years and the rounds that the data was found.

2013:
From February 14 to February 17, 144 golfers participated in the tournament, but 5 players did not compete in day 2 of the 4-day tournament. Thus, the data will be taken from 139 golfers.
Round 1
Hole 18- Par 4

X	1	2	3	4	5	6	7	8
P (X=x)	$\frac{0}{139}$	$\frac{0}{139}$	$\frac{12}{139}$	$\frac{90}{139}$	$\frac{32}{139}$	$\frac{3}{139}$	$\frac{2}{139}$	$\frac{0}{139}$

Following the work done to find the expected value with the 2012 data, the same process can be followed for upcoming years.



E (x)=4.23
σ=0.692

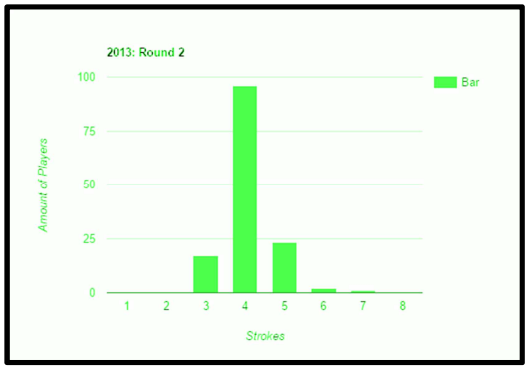
Analysis:
Within one standard deviation, the values are 3.538 and 4.922 in which $\frac{90}{139}$or 65% of the data is held. Two standard deviations holds values of 2.846 and 5.614 in which $\frac{134}{139}$or 96% of the data is contained. Three standard deviations have values of 2.184 and 6.306 in which $\frac{137}{139}$or 99% of the data is in range.

Reflection:
These values are not significantly off the ones of a normal distribution curve and hold efficacy to the scores representing the curve.

Round 2
Hole 18- Par 4

X	1	2	3	4	5	6	7	8
P (X=x)	$\frac{0}{139}$	$\frac{0}{139}$	$\frac{17}{139}$	$\frac{96}{139}$	$\frac{23}{139}$	$\frac{2}{139}$	$\frac{1}{139}$	$\frac{0}{139}$

E (x)= 4.09
σ=0.634



Analysis:
Within one standard deviation, the values are 3.456 and 4.724 in which $\frac{96}{139}$or 69% of the data is held. Two standard deviations holds values of 2.822 and 5.358 in which $\frac{136}{139}$or 98% of the data is contained. Three standard deviations have values of 2.188 and 5.987 where$\frac{136}{139}$or 98% of the data is still the range.

Reflection:
These values are fairly accurate to the ones of a normal distribution curve.

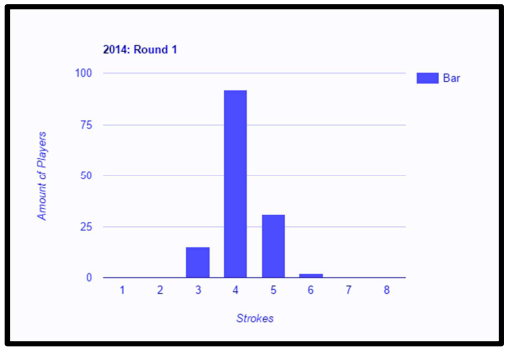
2014:

From February 13 to February 16, 142 golfers participated in the tournament, but 2 players did not compete in day 2 of the 4-day tournament. Thus, the data will be taken from 140 golfers.

Round 1
Hole 18- Par 4

X	1	2	3	4	5	6	7	8
P (X=x)	$\frac{0}{140}$	$\frac{0}{140}$	$\frac{15}{140}$	$\frac{92}{140}$	$\frac{31}{140}$	$\frac{2}{140}$	$\frac{0}{140}$	$\frac{0}{140}$

E (x)=4.14
σ=0.604



Analysis:
Within one standard deviation, the values would be 3.536 and 4.744 where$\frac{92}{140}$ or 66% of the data is held. Two standard deviation holds values of 2.932 and 5.348 in which $\frac{138}{140}$or 99% of the data is contained. Three standard deviations have values of 2.328 and 5.952 in

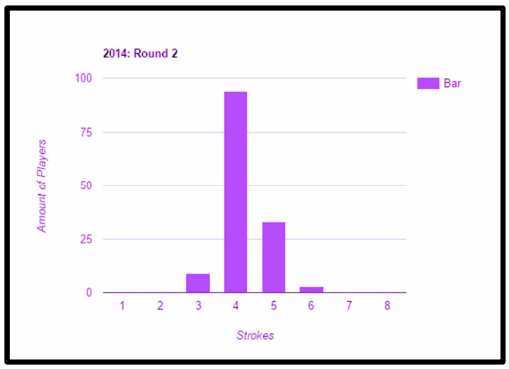
which $\frac{138}{140}$ or 99% of the data is in range.

Reflection:
These percentages, although not exactly alike, are not significantly different from normal distribution percentages.

Round 2
Hole 18- Par 4

X	1	2	3	4	5	6	7	8
P (X=x)	$\frac{0}{140}$	$\frac{1}{140}$	$\frac{9}{140}$	$\frac{94}{140}$	$\frac{33}{140}$	$\frac{3}{140}$	$\frac{0}{140}$	$\frac{0}{140}$

E (x)= 4.20
σ=0.612



Analysis:
Between one standard deviation, the values would be 3.588 and 4.812 where $\frac{94}{140}$ or 67% of the data is held. Two standard deviations holds values of 2.976 and 5.424 in which $\frac{136}{140}$ or 97% of the data is contained. Three standard deviations have values of 2.364 and 6.036 in which $\frac{139}{140}$ or 99% of the data is in range.

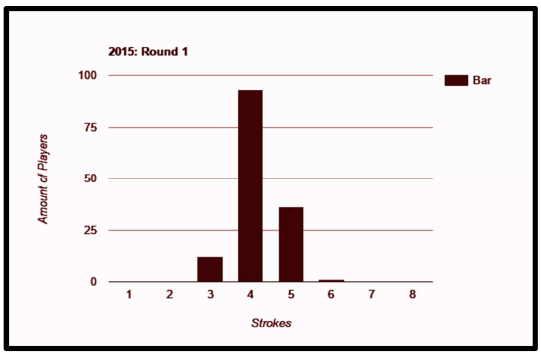
Reflection:
These percentages may not be congruent with the normal distribution percentages, but represent a similar result.

2015:
From February 19 to February 22, 144 golfers participated in this tournament, but 2 players did not compete in day 2 of the 4-day tournament. Thus, the data will be taken from 142 golfers.

Round 1
Hole 18- Par 4

X	1	2	3	4	5	6	7	8
P (X=x)	$\frac{0}{142}$	$\frac{0}{142}$	$\frac{12}{142}$	$\frac{93}{142}$	$\frac{36}{142}$	$\frac{1}{142}$	$\frac{0}{142}$	$\frac{0}{142}$

E (x)= 4.18
σ=0.577



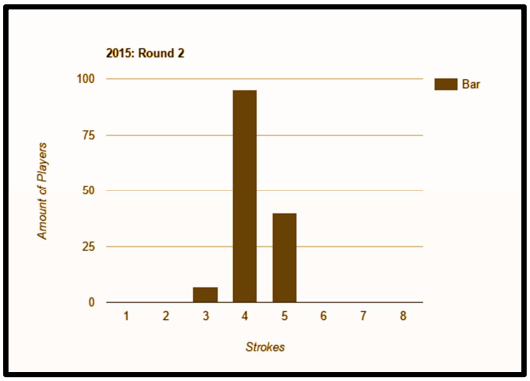
Analysis:
Within one standard deviation, the values are 3.603 and 4.757 where $\frac{93}{142}$ or 65% of the data is held. Between two standard deviations the values are 3.026 and 5.334 in which $\frac{129}{142}$ or 91% of the data is contained. Three standard deviations have values of 2.449 and 5.911 in which $\frac{141}{142}$ or 99% of the data is in range.

Reflection:
These percentages are not far off the ones attributed with normal distribution percentages.

Round 2
Hole 18- Par 4

X	1	2	3	4	5	6	7	8
P (X=x)	$\frac{0}{142}$	$\frac{0}{142}$	$\frac{7}{142}$	$\frac{95}{142}$	$\frac{40}{142}$	$\frac{0}{142}$	$\frac{0}{142}$	$\frac{0}{142}$

E (x)= 4.23
σ=0.526



Analysis:
Between one standard deviation, the values would be 3.704 and 4.756 where $\frac{95}{142}$ or 67% of the data is held. Two standard deviations holds values of 3.178 and 5.282 in which $\frac{135}{142}$ 95% of the data is contained. Three standard deviations have values of 2.652 and 5.808 in which $\frac{142}{142}$ or 100% of the data is in range.

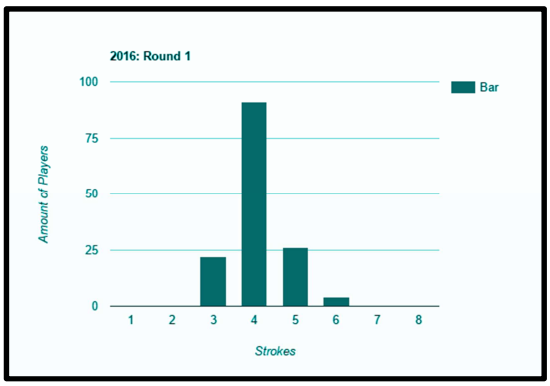
Reflection:
Essentially, these percentages are almost exactly like the ones that are typical to normal distribution percentages.

2016:
From February 18 to February 21, 145 golfers participated in this tournament, but 2 players did not compete in day 2 of the 4-day tournament. Thus, the data will be taken from 143 golfers.

Round 1
Hole 18- Par 4

X	1	2	3	4	5	6	7	8
P (X=x)	$\frac{0}{143}$	$\frac{0}{143}$	$\frac{22}{143}$	$\frac{91}{143}$	$\frac{26}{143}$	$\frac{4}{143}$	$\frac{0}{143}$	$\frac{0}{143}$

E (x)= 4.08
σ=0.664

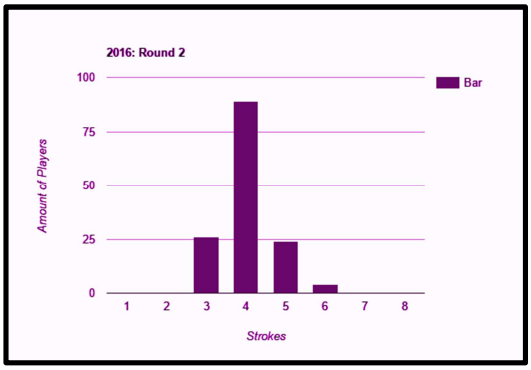


Analysis:
Between one standard deviation with values of 3.416 and 4.744 in which $\frac{91}{143}$ or 64% of the data is held. Two standard deviations holds values of 2.752 and 5.408 in which $\frac{139}{143}$ or 97% of the data is contained. Three standard deviations have values of 2.088 and 6.072 in which $\frac{143}{143}$ or 100% of the data is in range.

Reflection:
These percentages are fairly similar to the normal distribution curve percentages.

Round 2
Hole 18- Par 4

X	1	2	3	4	5	6	7	8
P (X=x)	$\frac{0}{143}$	$\frac{0}{143}$	$\frac{26}{143}$	$\frac{89}{143}$	$\frac{24}{143}$	$\frac{4}{143}$	$\frac{0}{143}$	$\frac{0}{143}$



E (x)= 4.04
σ=0.678

Analysis:
Between one standard deviation, the values are 3.362 and 4.718 where $\frac{89}{143}$ or 62% of the data is held. Two standard deviations holds values of 2.684 and 5.396 in which $\frac{139}{143}$ or 97% of the data is contained. Three standard deviations have values of 2.006 and 6.074 in which $\frac{143}{143}$ or 100% of the data is in range.

Reflection:
The percentages are not very close to the normal distribution, but give the same ideas present.

Prediction of 2017 Scores:

With the data collected from the previous past years, it is safe to assume that 2017 will have the same patterns. Therefore, I can assume that the mean of the golfers who participated will be 141 because I can use the number of golfers from each year and divide by the number of years: Mean=$\frac{141+139+140+142+143}{5}$= 141. Likewise, the mean of the Round 1 data for 1 stroke is 0. The mean of the Round 1 data for 2 strokes is Mean=$\frac{0+0+0+0+0}{5}$= 0. The mean of the Round 1 data for 3 strokes is Mean=$\frac{4+12+15+12+22}{5}$= 13. The mean of the Round 1 data for 4 strokes is 88. The mean of the Round 1 data for 5 strokes is 36. The mean of the Round 1 data for 6 strokes is 3.6, but in golf one cannot get a fraction of stroke, so rounding to 4 will be most reasonable. The mean of the Round 1

data for 7 strokes is 0.4. Again, this is not possible in golf; therefore rounding to 1 is most reasonable. Lastly, for Round 1 data for 8 stokes will be 0. The same math can be done for Round 2.

Round 1

Hole 18- Par 4

X	1	2	3	4	5	6	7	8
P (X=x)	0	0	13	88	36	4	1	0

E (x)= 4.24

σ= 0.681

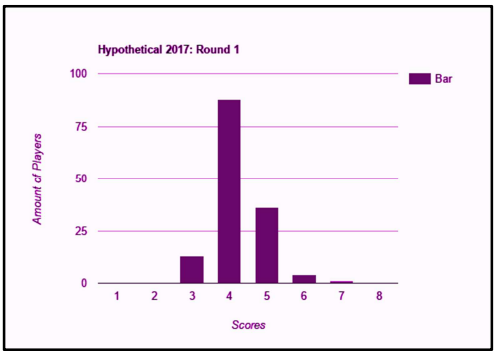
Round 2

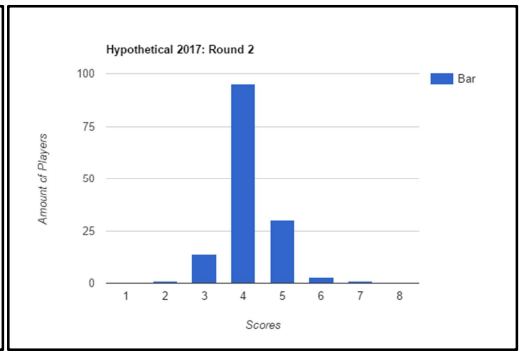
Hole 18- Par 4

X	1	2	3	4	5	6	7	8
P (X=x)	0	1	14	95	30	3	1	0

E (x)= 4.16

σ= 0.674





Actual Scores from 2017:

From February 16 to February 19, 144 golfers participated in this tournament, but 6 players did not complete day 2 of the 4-day tournament. Thus, the data will be taken from 138 golfers.

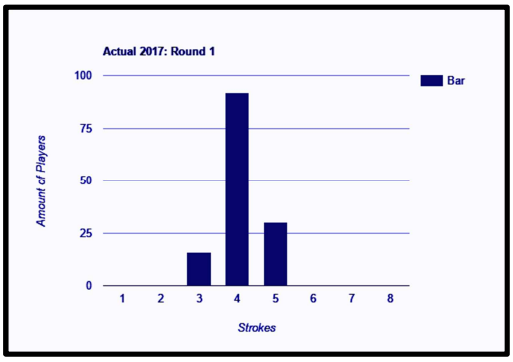
Round 1

Hole 18- Par 4

X	1	2	3	4	5	6	7	8
P (X=x)	$\frac{0}{138}$	$\frac{0}{138}$	$\frac{16}{138}$	$\frac{92}{138}$	$\frac{30}{138}$	$\frac{0}{138}$	$\frac{0}{138}$	$\frac{0}{138}$

E (x)=4.10

σ= 0.568



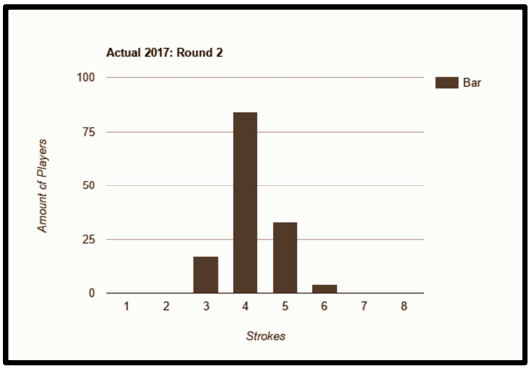
Round 2

Hole 18- Par 4

X	1	2	3	4	5	6	7	8
P (X=x)	$\frac{0}{138}$	$\frac{0}{138}$	$\frac{17}{138}$	$\frac{84}{138}$	$\frac{33}{138}$	$\frac{4}{138}$	$\frac{0}{138}$	$\frac{0}{138}$

E (x)= 4.17

σ= 0.669

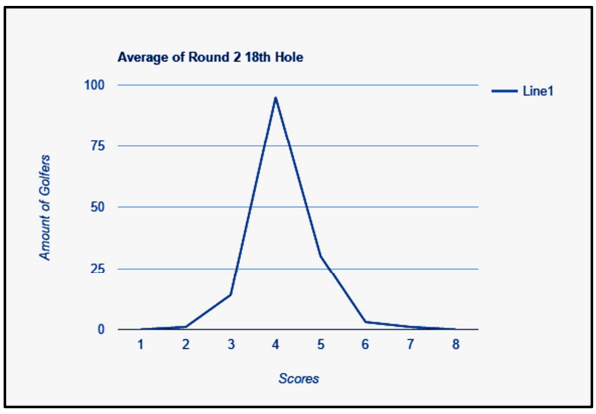


Analysis:

After gathering data from 2012 to 2016, Rounds 1 and 2, many similarities have been acknowledged. Initially, from looking at the tables, it can be inferred that most of the scores were better on Day 2. This could possibly be because golfers attending had become accustomed to the hole, the greens, the weather, and other attributes. This change created the shape of the curve to alter from Round 1. However, when visualized as a whole from the past 6 years of data from the Northern Trust Open, the data is not the typical normal distribution curve. Yet essentially, the data is a different form of, the bell shape curve that is usually attributed to normal distribution.

So, what deems normal distribution?

Data that is normally distributed has specific characteristics reckoning the curve as so. When data is distributed normally it skews heavily towards the central value with minimal bias to the left and right sides of the curve. This allows the "bell-curve" depiction. Additionally, the normally distributed data has a mean, median, and mode that are all equal. Likewise, there is symmetry in the center as 50% of the values are less and greater than the mean. Fundamentally, it is best suited for data that meets several different conditions like a strong tendency for the data to take a central value and frequency of the deviations must diminish while moving away from the central value. On the contrary, the data does have fairly have the same mean, median, and mode as most golfers shot par on the 18th hole at Riviera Country Club. The mean was not the same for every round as the frequency of the golfers getting the particular score on each hole differed every year; in fact it varied between days (Round 1 and 2). In golf, a player cannot get a decimal point added to their stroke. In this case, all of the means did not mean exactly 4 strokes. Essentially, it is stating that either 4 or 5 strokes would likely be taken for the ball to get to the hole.



As explained with the 2012 data and the other rounds of the years after, the standard deviation percentages were not exactly the values of

a normal distribution graph, but were fairly close. In that manner, the data collected is similar to the normal distribution percentages and therefore my investigation has been proved to have a positive result. The data is normally distributed. Though the data didn't fit the predetermined definition of a normal distribution, each bar graph illustrated the same sort of curve. The 2 and 3 strokes were typically more minimal than the 5 and 6 strokes taken on the hole. Therefore, the exemplary "normal" distribution is not patent, but a unique curve of normally distributed data is apparent. If the bar graph is looked at through the use of plotted points; the curve is not fully similar to a bell shape, yet depicts a resemblance to the regularly known curve. In that case, the data from the Northern Trust Open is technically normally distributed, but in a way that is distinctive to the data. The standard bell curve structure is solely changed a little, as both sides that trickle down are not quite analogous. So, in conclusion a normal distribution graph has been derived, but this particular one carries its own specific characteristics. Thus, a normal distribution curve can yield different characteristics, but the same intent, as is apparent in this example.

Conclusion:

As stated earlier, golf has always and will always be a part of me. Throughout the years I have discovered many important lessons through golf, but golf is not only an eye opener in situations, but also a great math investigation. Almost every yard of golf has some math involved in it, from calculating distances to measuring the slope of the green to the investigation of the normal distribution of golf scores on the 18th hole at Riviera Country Club. While gathering my data, I began exploring how course management picks a specific hole's par. Essentially; some golfers can play a par 5 as an easy par 4. Golfers such as Tiger Woods, Rory Mcilroy, Phil Mickelson, and other greats on the tour have even been drove their first shot onto the green from the tee box on par 4's. In that case, shouldn't it be a par 2? Although that would make it nearly impossible for me to shoot par or even bogey on the hole. Therefore, how is it chosen that the hole will be titled as the par it is given? After researching and talking to some of the coaches in my area; at their respective home courses, a multitude of players from different handicaps are first selected to play the course and the scores are then reviewed. The mean of all the numbers is then typically used to set the par for the hole. Evidently, this would explain why a majority of the professionals got to the hole in 4 strokes that meant that they ended their round with a par. To me, my initial hypothesis speculated this would happen, but to my surprise many professional golfers who have inched their way up to high rankings had some bogeys on the hole. That being said, this data broadened my horizons to think about other potential scenarios that can be inspected.

One function on my graphing calculator that greatly impressed me was the binompdf, binomcdf, normalpdf, and normalcdf buttons. Conceivably, I could use the distribution curve I found to help me with probabilities involved in golf. Specifically like how football fans have their fantasy league, I could have my own golf brackets. In this fantasy I could state that 35% of the golfers will make a birdie on one or more of the holes that they play on a particular day or I could research the likelihood that only the top 5% of the golfers on the tour will make the cut for Round 3 and 4. This could potentially go on forever, as there are many ways to take the distribution curve and see the probabilities of different outcomes. Another possibility would be to look at the golfer's score on the other par 4 holes to compare the distribution of scores and see if the other holes were perhaps more uniform to a bell shape. Likewise, if all 18 holes were investigated from the years 2012 to 2017, I could have a normal distribution arena and have impeccable chances at winning the golf bracket that I would create. Another exploration includes finding how many golfers shot birdie or better and using the probability to bet. Essentially, if a group of people wanted to bet on golf scores of the 18th hole because it is deemed the most important hole, one would have a better chance of being accurate with this prior knowledge.

All in all the data proved to be a unique shape of the normal distribution curve, even if the percentages attributed were not precisely the same. The shape altered because the nature of the

experiment. Golf data is discrete and not continuous. Likewise, there are no 3.5 or 5.1 scores in golf and so each bar graph was divided in the middle to be determined as a point on a graph. The points were then connected and like the graph where this was demonstrated, the form was nearly accurate. If it had been that more golfers got to the hole in 3 strokes or the opposite where less would have gotten to the hole in less than 5 strokes, the data would fit the bell shape better, but this cannot be controlled. Essentially, that is how the investigation grew. In essence, this investigation had a positive result and for future Northern Trust Opens (now Genesis Open), I will stay faithful that most golfers will reach the hole in 4 shots because I might even start putting down some money against the rest of my family.

The Future of the Normally Distributed 18th Hole Data:

As stated previously there are multiple ways to further use the data, one such way is using the normalcdf function on a calculator and finding the probability of a golfer scoring birdie or better on the 18th hole. Based on the years 2012-2016 "Prediction of 2017 Scores: Round 2" data**,** the probability of getting birdie or better is answered by using normalcdf (0, 3, 4.16, 0.674) = 0.0426 = 4.26%. This percentage is miniscule and so, if I were to potentially bet on the number of golfers scoring birdie or better, I would pick a lower number. This prediction could then be compared to the "Actual 2017: Round 2" data. There were 138 golfers who participated in the Northern Trust Open in Rounds 1 and Round 2; therefore, in all, there were 2(138)= 276 trials for getting a birdie or better. The probability will be the answer acquired above (0.0426). Lastly, when choosing the value for how many scores will be birdie or better, I predict that 20 scores will be birdie or better. Thus, the function would read: binompdf (276, 0.0426, 20) = 0.00748 = 0.7% This value is extremely low; if there was only less than a 1% chance to win money if I were to bet, I might put my money down based on particular conditions. For example, if a situation were posed where I have to bet one dollar to win a prize, the equation would read:

T= amount of money if I won

$E(x) = (0.00748 \times T) + ((1-0.00748) \times (-1)) > 0$

$0.00748T - 0.99252 > 0$

$0.00748T > 0.99252$

$T > 133$

From the inequality above, if the prize money (T) is more than $133, I will be winning money down the line. Hence, in a situation where I make 1,000 bets on how many scores will be birdie or better, with the 0.7% chance that it is what the score will be, I will potentially lose 993 bets and win 7 times. Furthering the example above, if the value of the prize money (T) is $10,000 and I bet 1,000 times putting $1 down, I will lose $983, but win 7 times, which would mean winning $70,000 subtracting the $993 from the bets I lost. Now, if the T= $10,000 is plugged into the calculation, I will roughly be making ((0.00748)(10,000) + (1-0.00748)(-1)) = about $73.8 earnings each bet. However, if the prize money (T) value is exactly $133, this is the breakeven point where I will not win money and if the value of T were less than $133, I would lose money.

Reflection:

Before this extension of my IA, I lacked knowledge about how betting works. To me, I believed the probability was the most important part of winning; however, it is the involvement of probability weighted on expected value. In the future, I could analyze more profitable and unprofitable bets.

As aforementioned, math can be found everywhere in golf, thus there are still many other ways to explore the world of math and golf.

IB Mathematics SL: Exploration Feedback to student			Name:	
Title:	Normal Distribution of the 18th Hole	Date Set: March 24, 2017		Date Submitted: March 24, 2017

A Communication	Score:	4	marks (out of 4 possible marks)

- Expressed ideas clearly
- Identified a clear aim for the exploration
- Focused on the aim and avoiding irrelevance
- Structured ideas in a logical manner
- Edited the exploration so that it is easy to follow
- Cited references where appropriate
- Included Introduction, rationale, description of aims, conclusion
- Graphs, tables and diagrams accompanied the work in the appropriate place and were not attached as appendices to the document.

4 The exploration is coherent, well organized, concise and complete.

Comments: Student expresses ideas clearly and identifies aim, plan, analysis, reflection, and conclusion throughout the exploration. Her graphs and tables were appropriate throughout the course. Her labeling made the exploration easy to follow.

B Mathematical presentation	Score:	3	marks (out of 3 possible marks)

- Used appropriate mathematical language and representation (notation, symbols, terminology)
- Defined key terms, where required
- Selected appropriate mathematical tools (including information and communication technology)
- Expressed results to an appropriate degree of accuracy
- Used multiple forms of mathematical representation, such as formulae, diagrams, tables, charts, graphs and models, where appropriate.
- Used mathematical language when communicating mathematical ideas, reasoning and findings.
- Chose and used appropriate ICT tools such as graphic display calculators, screenshots, graphing, spreadsheets, databases, drawing and word-processing software, as appropriate, to enhance mathematical communication.

3 The mathematical presentation is appropriate throughout.

Comments: Uses multiple representations of all of her data. Results are accurate and explain key terms regarding golf and mathematics.

C Personal engagement	Score:	4	marks (out of 4 possible marks)

- Worked independently
- Asked questions, making conjectures and investigating mathematical ideas
- Read mathematics and researched areas of interest
- Looked for and creating mathematical models for real-world situations
- Considered historical and global perspectives
- Demonstrated that he or she has "made the exploration their own"
- Expressed ideas in an individual way
- The extent to which the student engaged with the exploration and made it their own.
- Thought independently and/or creatively
- Addressed personal interest
- Presented mathematical ideas in their own way.
- Evidence that the student made the work their own, not just simply a written report.
- Asked and answered questions: "I wonder if…", "What would happen if…" "Why does that happen…"
- The student's voice could be heard

4 There is abundant evidence of outstanding personal engagement.

Comments: Student's exploration is genuinely authentic. Her personal connection to golf and aim to model a real life golf situation with a normal distribution. Can clearly hear student's voice. (ex. "to me, my intial hypothesis speculated this would happen, but to my surprise…". Student displayed creativity to think of modeling with a normal distribution and then verifying the validity of her model.

D Reflection	Score:	2	marks (out of 3 possible marks)

- How the student reviewed, analyzed and evaluated the exploration.
- Discussed the implications of results
- Considered the significance of the exploration
- Looked at possible limitations and/or extensions
- Made links to different fields and/or areas of mathematics

2 There is evidence of meaningful reflection.

Comments: Her reflection was substantial in her conclusion, especially when she looked into the betting aspect as an application to her aim. However, her reflection throughout the paper was relatively superficial.

E Use of mathematics	Score:	6	marks (out of 6 possible marks)

- Demonstrated knowledge and understanding
- Produced work that is within the level of the course. It should not be solely based on mathematics listed in the prior learning.
- Applied mathematics in different contexts
- Applied problem-solving techniques
- Recognized and explained patterns, where appropriate
- This criterion assesses to what extent students used mathematics in the exploration.
- Sophistication in mathematics may include understanding and use of challenging mathematical concepts, looking at a problem from different perspectives and seeing underlying structures to link different areas of mathematics.
- Rigour involves clarity of logic and language when making mathematical arguments and calculations.
- Precise mathematics is error-free and uses an appropriate level of accuracy at all times.

6 Relevant mathematics commensurate with the level of the course is used. The mathematics explored is correct. Thorough knowledge and understanding are demonstrated.

Comments: She demonstrates a thorough understanding of normal distribution and it's real world application. She conceptually grasps the concept of standard deviation and expected value. Her knowledge is evident when she connects the interplay of probability and pay off in expected value.

Total Marks Earned:	19	marks (out of 20 possible marks)